1) a)

$x_0 = 1, y_0 = 2$

$y_1 = 2 + 0.5\mathrm{f}(x_0, y_0)$

$= 2 + 0.5 \times 0 = 2$

$x_1 = 1 + 0.5 = 1.5$

$\therefore y(2) = 2$

1) b)

$y_2 = y_0 + 2h\mathrm{f}(x_1, y_1)$

$= 2 + 2 \times 0.5 \times 0.8047189562 \ldots = 2.804718956 \ldots$

$x_2 = 1.5 + 0.5 = 2$

$\therefore y(2) = 2.805$ (3 DP)

2)

$x_0 = 2, y_0 = 1$

$y_1 = 1 + 0.1\mathrm{f}(x_0, y_0)$

$= 1 + 0.1 \times 2.693147181 \ldots = 1.269314718 \ldots$

$x_1 = 2 + 0.1 = 2.1$

$y_2 = y_0 + 2h\mathrm{f}(x_1, y_1)$

$= 1 + 2 \times 0.1 \times 2.9194779 \ldots = 1.58389558 \ldots$

$x_2 = 2.1 + 0.1 = 2.2$

$\therefore y(2.2) = 1.584$ (3 DP)

3) a)

$x_0 = 0, y_0 = 1$

$x_1 = 0.75, y_1 = 1.75$

$y_2 = y_0 + 2h\mathrm{f}(x_1, y_1)$

$= 1 + 2 \times 0.75 \times 2.943137453 \ldots = 5.41470618 \ldots$

$\therefore y(1.5) = 5.415$ (3 DP)

3) b)

$x_0 = 1, y_0 = 0.2$

$x_1 = 1.25, y_1 = 0.2779$

$y_2 = y_0 + 2hf(x_1, y_1)$

$= 0.2 + 2 \times 0.25 \times 0.8363594124 \ldots = 0.6181797062 \ldots$

$x_2 = 1.25 + 0.25 = 1.5$

$y_3 = y_1 + 2hf(x_2, y_2)$

$= 0.2779 + 2 \times 0.25 \times 8.71721164 \ldots = 4.63650582 \ldots$

$\therefore y(1.75) = 4.637$ (3 DP)

4) a)

$x_0 = 1, y_0 = 3$

$k_1 = 2.633916794 \ldots$

$k_2 = 4.2501647216535 \ldots$

$x_1 = 3, y_1 = 3 + \frac{1}{2}(2.633916794 \ldots + 4.2501647216535 \ldots) = 6.442040258 \ldots$

$\therefore y(3) = 6.44$

4) b)

$x_0 = 2, y_0 = 0.5$

$k_1 = 0.2459397694 \ldots$

$k_2 = 0.3321224914 \ldots$

$x_1 = 2.2, y_1 = 0.5 + \frac{1}{2}(0.2459397694 \ldots + 0.3321224914 \ldots) = 0.7890461304 \ldots$

$\therefore y(2.2) = 0.789$

4) c)

$x_0 = 0, y_0 = 3$

$k_1 = 1.732050808 \ldots$

$k_2 = 4.002893403 \ldots$

$x_1 = 3, y_1 = 3 + \frac{1}{2}(1.732050808 \ldots + 4.002893403 \ldots) = 5.867472105 \ldots$

$\therefore y(0.5) = 5.87$

5) a)

$x_0 = 4,\ y_0 = \mathrm{e}^{0.1}$

$y_1 = \mathrm{e}^{0.1} + 0.1\mathrm{f}(x_0, y_0)$

$= \mathrm{e}^{0.1} + 0.1 \times 3.315512754 \ldots = 1.436722193 \ldots$

$x_1 = 4 + 0.1 = 4.1$

$y_2 = y_0 + 2h\mathrm{f}(x_1, y_1)$

$= \mathrm{e}^{0.1} + 2 \times 0.1 \times 4.357794176 \ldots = 1.97672953 \ldots$

$x_2 = 4.1 + 0.1 = 4.2$

$\therefore y(4.2) = 1.977$ (3 SF)

5) b)

$\int \frac{1}{y}\mathrm{d}y = \int \sqrt{2x+1}\,\mathrm{d}x = \int (2x+1)^{\frac{1}{2}}\mathrm{d}x$

$\ln y = \frac{1}{3}(2x+1)^{\frac{3}{2}} + c$

Given $(4,\ \mathrm{e}^{0.1})$ then

$\ln(\mathrm{e}^{0.1}) = \frac{1}{3}(2 \times 4 + 1)^{\frac{3}{2}} + c \Rightarrow c = -8.9$

$\therefore \ln y = \frac{1}{3}(2x+1)^{\frac{3}{2}} - 8.9$

5) c)

$$\frac{1.977 - \mathrm{e}^{\frac{1}{3}(2\times 4.2+1)^{\frac{3}{2}} - 8.9}}{\mathrm{e}^{\frac{1}{3}(2\times 4.2+1)^{\frac{3}{2}} - 8.9}} = -0.025$$

6) a)

$x_0 = 0,\ y_0 = 1$

$y_1 = 1 + 0.05\mathrm{f}(x_0, y_0)$

$= 1 + 0.05 \times -\frac{1}{2} = 0.975$

$x_1 = 0 + 0.05 = 0.05$

$y_2 = y_0 + 2h\mathrm{f}(x_1, y_1)$

$= 1 + 2 \times 0.05 \times -0.4375 = 0.95625$

$x_2 = 0.05 + 0.05 = 0.1$

$\therefore y(0.1) = 0.95625$

6) b)

$\frac{dy}{dx} + \frac{1}{2}y = x$

I.F. is $e^{\int \frac{1}{2} dx} = e^{\frac{x}{2}}$

$e^{\frac{x}{2}} \frac{dy}{dx} + \frac{1}{2} e^{\frac{x}{2}} y = x e^{\frac{x}{2}}$

$e^{\frac{x}{2}} y = \int x e^{\frac{x}{2}} dx$

Use Integration by Parts:

$u = x$ $\quad$ $\frac{dv}{dx} = e^{\frac{x}{2}}$

$\frac{du}{dx} = 1$ $\quad$ $v = 2e^{\frac{x}{2}}$

$\therefore \int x e^{\frac{x}{2}} dx = 2x e^{\frac{x}{2}} - \int 2e^{\frac{x}{2}} dx = 2x e^{\frac{x}{2}} - 4e^{\frac{x}{2}} (+c)$

So that:

$e^{\frac{x}{2}} y = 2x e^{\frac{x}{2}} - 4e^{\frac{x}{2}} + c$

$y = 2x - 4 + ce^{-\frac{x}{2}}$

6) c)

Given $(0, 1)$ then

$1 = 0 - 4 + c \Rightarrow c = 5$

$\therefore y = 2x - 4 + 5e^{-\frac{x}{2}}$

When $x = 0.1$, $y = 2 \times 0.1 - 4 + 5e^{-\frac{0.1}{2}} = 0.956147 \ldots$

$\therefore y(0.1) = 0.956$ (3 SF)

6) d)

The value from part a) is equal to the answer in part c) to three significant figures, meaning the estimate in part a) is very accurate.

7) a)

$\frac{dC}{dt} = -0.2C + 1.5e^{-0.1t}$

$t_0 = 0$, $C_0 = 0$

$C_1 = 0 + 0.5\text{f}(t_0, C_0)$

$= 0 + 0.5 \times 1.5 = 0.75$

$\therefore C = 0.75$ mg/L

7) b)

A greater number of smaller steps could be used.

7) c)

$t_0 = 0,\ C_0 = 0$

$k_1 = hf(x_r, y_r) = 0.5f(t_0, C_0) = 0.5 \times 1.5 = 0.75$

$k_2 = hf(x_r + h, y_r + k_1) = 0.5f(0.5, 0.75) = 0.6384220684\ \ldots$

$y_{r+1} = y_r + \frac{1}{2}(k_1 + k_2)$

$C_1 = 0 + \frac{1}{2}(0.75 + 0.6384220684\ \ldots)$

$= 0.6942110342\ \ldots$

$t_1 = 0 + 0.5 = 0.5$

$\therefore C = 0.69$ mg/L

8) a)

$(15 - 0.04t^2)\frac{dT}{dt} = 6t - 0.8T \Rightarrow \frac{dT}{dt} = \frac{6t - 0.8T}{15 - 0.04t^2}$

$t_0 = 0,\ T_0 = 20$

$T_1 = 20 + 0.5f(t_0, T_0)$

$= 20 + 0.5 \times -\frac{16}{15} = 19.46666667\ \ldots$

$t_1 = 0 + 0.5 = 0.5$

$T_2 = 19.46666667\ \ldots + 0.5f(t_1, T_1)$

$= 19.46666667\ \ldots + 0.5 \times -0.8387814098\ \ldots = 19.04727596$

$t_2 = 0.5 + 0.5 = 1$

$\therefore T = 19.0°C$

8) b)

A greater number of smaller steps could be used.

8) c)

$t_0 = 0, T_0 = 20$

$t_0 + \frac{h}{2} = 0 + \frac{1}{2} = 0.5$

$T_0 + \frac{h}{2}\mathrm{f}(t_0, T_0) = 20 + \frac{1}{2} \times -\frac{16}{15} = 19.46666667 \ldots$

$\therefore\ T_1 = T_0 + h\mathrm{f}\left(t_0 + \frac{h}{2}, T_0 + \frac{h}{2}\mathrm{f}(t_0, T_0)\right)$

$= 20\ + 1 \times \mathrm{f}(0.5, 19.46666667 \ldots)$

$= 20\ + 1 \times -0.8382222222 \ldots$

$= 19.161777777 \ldots$

$\therefore T = 19.2°\mathrm{C}$

9)

$x_0 = 0, y_0 = 1$

$k_1 = \mathrm{f}(x_0, y_0) = \mathrm{f}(0, 1) = 0 - 1 = -1$

$k_2 = \mathrm{f}(x_0 + 0.25, y_0 + 0.25 \times -1) = \mathrm{f}(0.25, 0.75) = 0.25 - 0.75 = -0.5$

$k_3 = \mathrm{f}(x_0 + 0.25, y_0 + 0.25 \times -0.5) = \mathrm{f}(0.25, 0.875) = -0.625$

$k_4 = \mathrm{f}(x_0 + 0.5, y_0 + 0.5 \times -0.625) = \mathrm{f}(0.5, 0.6875) = -0.1875$

$y_1 = 1 + \frac{1}{12}(-1 + 2(-0.5) + 2(-0.625) - 0.1875) = 0.713541666 \ldots$

$\therefore y = 0.714$ (3 DP)

10)

$h = 0.1$

$x_0 = 1, y_0 = a$

$y_1 = a + 0.1\mathrm{f}(x_0, y_0) = a + 0.1 \times (5 - a) = 0.9a + 0.5$

$x_1 = 1 + 0.1 = 1.1$

$y_2 = y_0 + 2h\mathrm{f}(x_1, y_1) = a + 0.2 \times \big(5.5 - (0.9a + 0.5)\big) = 0.82a + 1$

$\Rightarrow\ 0.82a + 1 = 3.05$

$\therefore a = 2.5$

11) a)

$x_0 = 1, y_0 = 0.5$

$y_1 = 0.5 + h\mathrm{f}(x_0, y_0) = 0.5 + h \times (1 - 0.5) = 0.5h + 0.5$

Since $y_1 = 0.6$ then $0.5h + 0.5 = 0.6$

$\therefore h = 0.2$

11) b)

$x_1 = 1 + 0.2 = 1.2$

$y_2 = y_0 + 2hf(x_1, y_1) = 0.5 + 2 \times 0.2(1.2^2 - 0.6) = 0.836$

$x_2 = 1.2 + 0.2 = 1.4$

$\therefore (x_1, y_1) = (1.4, 0.836)$

12)

$x_0 = 0, y_0 = 1$

$y_1 = 1 + hf(x_0, y_0) = 1 + h \times (0 + 1) = 1 + h$

$x_1 = 0 + h = h$

$y_2 = 1 + 2hf(x_1, y_1) = 1 + 2h \times (1 + h + h) = 4h^2 + 2h + 1$

Since $y_2 = 1.96$ then $4h^2 + 2h + 1 = 1.96 \Rightarrow 4h^2 + 2h - 0.96 = 0$

$\Rightarrow h = -0.8$ or $h = 0.3$

$\therefore h = 0.3$

13)

$h = 0.5$

$x_0 = 2, y_0 = a$

$k_1 = 0.5(2a - 6) = a - 3$

$k_2 = 0.5\big(2(a + a - 3) - 3(2 + 0.5)\big) = 2a - 6.75$

$y_1 = a + \frac{1}{2}(a - 3 + 2a - 6.75) = 2.5a - 4.875$

$\therefore y(2.5) = 7.625 \Rightarrow 2.5a - 4.875 = 7.625$

$\therefore a = 5$

14)

$x_0 = 0, y_0 = 1$

$k_1 = h(4 + 0) = 4h$

$k_2 = h(4(1 + 4h) + 2h) = 18h^2 + 4h$

$y_1 = 1 + \frac{1}{2}(4h + 18h^2 + 4h)$

$\therefore 1 + \frac{1}{2}(4h + 18h^2 + 4h) = 2.16 \Rightarrow 9h^2 + 4h - 1.16 = 0$

$\Rightarrow h = -0.6444444 \ldots$ or $h = 0.2$

$\therefore h = 0.2$